

Summer Funding Plan

I intend to use the Hamilton Award to enable full-time commitment to research this summer. Doing so will allow me to focus the entirety of my energies to research, speeding progress by enabling me to advance my research goals more quickly.

Tissues and individual cells are poroelastic, consisting of a background matrix suffused by a solvent carrying solutes of varying sizes. In vivo, materials undergo dynamic stresses, causing deformations which induce advection and size-dependent transport. Solute distribution in the nucleus, a heterogeneous poroelastic environment often subjected to dynamic strains, drives gene transcription. Experiments examining the macroscale dynamics of poroelastic media demonstrate enhanced concentration of large solutes in cyclically strained hydrogels, but there remains a dearth of studies probing the intricate interactions between solutes, pores, stress gradients, and microfluidics during dynamic strains, thus the roots of these emergent behaviors remain unclear. Diffusive transport of large solutes in poroelastic media is anomalous, strain-dependent, and temporally anisotropic, necessitating high spatiotemporal resolution.

To date, my work has provided the first nanoscopic, single-particle insight into the dynamics of these system. Both static and dynamic strains increase long-term localization and decrease diffusivity compared to unstrained controls. Strains also enhances the magnitude and duration of non-Gaussian diffusion. Dynamic strain induces asymmetric advective transport of large solutes during compression and relaxation, while also reducing long-term solute localization asymmetrically throughout their cycle by as much as threefold compared to static strains.

Over the course of the summer, I will apply strains to cells to understand how strains impact the transport of transcription factors *in situ*. To accomplish this goal, individual fluorescently labelled transcription factors must be imaged in a live cellular nuclei being subjected to well-characterized strains.

To accomplish this, human cells will be cultured by collaborators with extensive experience. These cells will be treated with a non-toxic HALO-tag solution to fluorescently label transcription factors (TFs) within the cell nuclei. Our microscope operates with low phototoxicity inside a climate-controlled incubation system, allowing cell health to be maintained for extended experimental sessions. Precise strain will be applied during live imaging using a spherical indenter mounted on a piezo-actuated atomic force microscope cantilever. This strategy was developed and successfully applied in similar prior experiments involving a poroelastic gel, and should be adaptable to this experiment with minimal alteration.

High speed imaging fluorescently-labelled TFs will be conducted using light sheet fluorescence microscopy to acquire high sensitivity, high-framerate time series of the particles. The necessary microscopy techniques were developed and in the course of previous work; this system has imaged quantum dots at up to 800fps, and this task should be well within demonstrated capabilities. Particle tracking will be conducted using TrackMate, open-source state-of-the-art particle tracking software, and the data will be analyzed using MATLAB scripts. This experiment is likely to be very similar to a recent collaboration tracking HALO-tagged histone protein complexes in the cellular nuclei, and the lessons learned there can be applied here.

As an extension, a unique in-house instrument which enables simultaneous atomic force microscopy and light sheet fluorescence microscopy could also be used to produce simultaneous particle tracking and rheological data to examine diffusive dynamics as a function of strain.

Alexander Marshall | Physicist

1900 Glenacre Court | Kernersville, North Carolina | 27284

☎ (336) 618-9851 • ✉ AlexanderNMarshall@proton.me

🌐 philosophiaephysica.org

Royster Fellow | Doctoral Candidate | UNC-Chapel Hill

Education

Doctorate of Philosophy in Physics

Chapel Hill, NC

The University of North Carolina at Chapel Hill

2027

Royster Doctoral Fellow | Exams Passed | All But Dissertation

Master of Science in Physics

Chapel Hill, NC

The University of North Carolina at Chapel Hill

2025

Completed *en route* to PhD

Physics B.S. | Philosophy B.A. | Mathematics Minor

Winston-Salem, NC

Wake Forest University

2022

GPA 3.91 | *Summa Cum Laude* | Honors in Physics | Stamps Scholar

Academic Projects

Doctoral Thesis

Anomalous Diffusion in Poroelastic Media

Advisor: Richard Superfine

Thesis Committee: Dr. Richard Superfine, Dr. C.A. Forrest, Dr. David Hill, Dr. Amy Oldenburg, Dr. Ehssan Nazockdast

Description: Explore the role of static and dynamic strains in the transport of large solutes in various poroelastic media, including transcription factor search patterns in cellular nuclei exposed to dynamic strains.

Masters Thesis

Physiological Strains in Poroelastic Media Induce Asymmetric Transport

Advisor: Richard Superfine

Thesis Committee: Dr. Richard Superfine, Dr. Greg M. Forrest, Dr. David Hill, Dr. Amy Oldenburg, Dr. Kerry Bloom

Description: Tracked quantum dots undergoing dynamic diffusion in poroelastic media using LSFM with unprecedented temporal resolution. Live imaging during physiological strain reveals that large solute transport is asymmetric during dynamic strains, with advection yielding significantly more forward angular anisotropy during the relaxation stroke than the compression stroke. These results suggest single-particle effects could explain the macroscopic solute concentration enhancements reported by studies of poroelastic media under dynamic strain.

Undergraduate Honors Thesis.....

Generating a 3-D Lattice of Photonic Spheres to Track Fluorescently Labelled Chromatin

Advisor: Dr. Keith Bonin

Committee: Dr. Keith Bonin, Dr. George Holzwarth, Dr. Stephen Baker

Description: Wrote a custom iterative routine to generate, test, and analyze hundreds of thousands of phase settings on a spatial light modulator. Used this routine to create 3D lattice of photonic spheres to be used for tracking labelled chromatin in irradiated cells.

Additional Research.....

Chromatin Structural Evolution Alters Histone Transport Dynamics **Chapel Hill, NC**

UNC-Chapel Hill, Dr. Kerry Bloom

2025

Tracking and analyzing the trajectories of HALO-tagged histones during metaphase and interphase to probe structural changes occurring in chromatin.

Enabling Low-Resource Microscopy

Chapel Hill, NC

UNC-Chapel Hill, Dr. Richard Superfine

2023-Present

Supervised a primarily undergraduate-driven effort to develop cutting edge microscopy techniques for use in low-research environments. Undergraduates won prizes at annual UNC MakerFest competition for two consecutive years.

Employing Advanced LSFM for Enhanced Imaging

Chapel Hill, NC

UNC-Chapel Hill, Dr. Richard Superfine

2022-Present

Engineered an advanced optical system to conduct new methods of LSFM, such as HILO and HIST. Due to software rot, had to rebuild system from the ground up, involving extensive design and engineering. Also implemented new optical modules and DAQ-board mediated software to enable sub-millisecond automated control of optical hardware.

Synthesizing Chalcogenide Crystals for Quantum Computing

Winston-Salem, NC

Wake Forest Nanotech Center, Dr. David Carroll

2021

Assisted with the synthesis of chalcogenide crystals being investigated as a potential substrate for quantum computing.

Neurochemical Addiction Mechanisms In the Nucleus Accumbens

Winston-Salem, NC

Wake Forest School of Medicine, Dr. Evgeny Budygin

2019-2021

Conducted cyclic voltammetry on rats *in vivo* and mouse brain slices *in vitro* to explore the role of neurotransmitters in addictive behaviors.

Cultural Impact of Oenology in the Peloponnese

Winston-Salem, NC

Wake Forest Department of Anthropology, Dr. Karen Friederic

2019

Conducted a month-long anthropological study in Nemea, Greece on how the economic significance of the viticultural industry impacts local culture. I found that extreme seasonal workloads intersect with traditional gendered domestic labor expectations to inhibit female career growth in this industry.

Cancer and Postoperative Pain

Winston-Salem, NC

Wake Forest Baptist Medical Center, Dr. Christopher Peters

2018

Prepared solutions, assisted with operations on research subjects, and conducted fluorescence microscopy on brain slices from rats to ascertain the distribution of optogenetic proteins in neurons after treatment with a viral vector.

Scholarships and Grants

Royster Doctoral Fellowship

Royster Foundation, UNC-Chapel Hill Graduate School

Five-year doctoral recruitment fellowship. Provides extradepartmental funding for the majority of a five year doctoral program.

Chapel Hill, NC

2022-2027

Stamps Scholarship

Stamps Foundation

Four-year scholarship awarded for academic merit. Covered all costs of attendance to Wake Forest University, including room and board. Also provided a technology grant and semesterly stipend.

Chapel Hill, NC

2018-2022

Summer Research Project Grant

Wake Forest University

Granted for summer anthropology project studying the cultural impact of viticulture in the Peleponnese.

Winston-Salem, NC

2019

Publications

Strain alters large solute transport in poroelastic media

Manuscript In Progress, UNC-Chapel Hill

Alexander Marshall, Richard Superfine

Chapel Hill, NC

Anticip. 2026

Dynamic strains drive asymmetric advection in poroelastic media

Manuscript In Progress, UNC-Chapel Hill

Alexander Marshall, Richard Superfine

Chapel Hill, NC

Anticip. 2026

Technical skills

Laboratory Safety Manager

UNC-Chapel Hill, Dept. of Environmental Health and Safety

Responsibilities include writing the lab safety plan, ensuring compliance during day-to-day operations, ensuring safety training compliance, disposing of hazardous waste, coordinating with EHS to maintain a safe workplace, and conducting regular inspections of critical safety equipment.

Chapel Hill, NC

2022-Present

Programming: MATLAB, LabVIEW, Python, L^AT_EX, Git, Arduino

Software: Microsoft Office, Spot-On, μ Manager, ImageJ, FIJI, autoCAD

Data Analysis: Particle Tracking, Image Analysis, Stochastic Processes, Machine Vision, Image Deconvolution

Microscopy: Transmission Microscopy, Fluorescent Microscopy, Light Sheet Fluorescent Microscopy, Total Internal Reflection Microscopy, Confocal Microscopy

Optical Systems: Custom Optical Design, Laser Systems, Spatial Light Modulation, Remote Focusing, Accusto-Optic Tunable Filters, 4-F System Design

Electronics: Multimeter proficiency, soldering, custom electronics design, multimodal control design, system timing control, DAQ-board interfacing, system control software design, Arduino-based control, Raspberry-Pi based control

General Lab Proficiencies: Chemical protocols, highly toxic chemical safety, fume hoods, cell culture, power tool use, general workshop skills

Machine Shop: Took a semester-long machine shop course at Wake Forest University in 2021.

Certified as departmental machine shop operator following this course.

Talks and Symposia

Physiological Strains in Poroelastic Media Induce Asymmetric Transport <i>Master's Defense, UNC-Chapel Hill</i>	Chapel Hill, NC November 17, 2025
Strain Alters Anomalous Diffusion in Poroelastic Media. <i>Symposium Talk, Elon University</i>	Elon, NC October 1, 2025
Diffusion in Strained Poroelastic Media <i>Thesis Proposal, UNC-Chapel Hill</i>	Chapel Hill, NC September 16, 2025
Generating a 3-D Lattice of Photonic Spheres <i>Honors Thesis Defense, Wake Forest University</i>	Winston-Salem, NC May 2022
Generating a 3-D Lattice of Photonic Spheres <i>Symposium Talk, Wake Forest University</i>	Winston-Salem, NC May 2022

Conferences

Strain-Driven Transport In Poroelastic Media <i>Biophysical Society Meeting, Biophysical Society</i>	San Francisco, CA February 21, 2026
Strain-Driven Transport In Poroelastic Media <i>13th Carolina Biophysics Symposium, UNC-Chapel Hill</i>	Chapel Hill, NC November 6, 2025
Strain Driven Transport In Poroelastic Media <i>Triangle Cytoskeleton Meeting, Duke University</i>	Durham, NC September 15, 2025
Strain-Driven Transport In Poroelastic Media <i>Royster Global Conference, Royster Society of Fellows</i>	Chapel Hill, NC October 21, 2025
Strain-Driven Transport In Poroelastic Media <i>Triangle Soft Matter Workshop, Duke University Dept. of Physics</i>	Durham, NC May 12, 2025
Strain-Driven Transport In Poroelastic Media <i>8th International Soft Matter Conference, Soft Matter Association of the Americas</i>	Raleigh, NC August 2024
Strain Driven Transport In Poroelastic Media <i>Triangle Cytoskeleton Meeting</i>	Raleigh, NC September 23, 2024
Probing Structural and Force Dynamics during Phagocytosis <i>Triangle Cytoskeleton Meeting</i>	Raleigh, NC September 25, 2023
Various topics <i>Annual Colloquium, UNC Applied Physical Sciences Department</i>	Chapel Hill, NC Annually since 2022
Stemps Scholars Convention <i>Attendee, University of Georgia</i>	Atlanta, Georgia Fall 2019
The Cultural Impact of Oenology in the Peleponnese <i>Undergraduate Research Day, Wake Forest University URECA Center</i>	Winston-Salem, NC September 2019

Teaching

Research Mentor

Superfine Lab, UNC-Chapel Hill

Chapel Hill, NC

Fall 2022-Present

Mentored a diverse group of undergraduates, helping them to develop as researchers while working on various projects. Persistent efforts at outreach drove undergraduate recruitment and engagements in the lab, leading to a dramatic expansion of the lab's engagement with undergraduates. The research activities of select students I advised are summarized in the mentorship section.

Teaching Assistant

Physics I, Wake Forest University

Winston-Salem, NC

Fall 2021

Due to COVID-19 restrictions, the instructor of record was unable to acquire a work visa and could not enter the United States, requiring the assumption of additional responsibilities.

Teaching Assistant

Physics I, UNC-Chapel Hill

Chapel Hill, NC

2023-2024

Primarily acted as a lab instructor across three semesters. Responsible for day-to-day administration of two 110 minute 75 student lab sections which met biweekly.

Teaching Assistant

Various Physics Courses, UNC-Chapel Hill

Chapel Hill, NC

2023-2024

Served as a TA for various other courses. Topics covered included Thermodynamics, Quantum Physics II, a first-year seminar centered on the philosophy of science, working in the departmental assistance center, and evaluating student performance.

Departmental Tutor

Physics I and Physics II, UNC-Chapel Hill Naval Armory

Chapel Hill, NC

Fall 2023-Spring 2024

Provided physics tutoring for ROTC-enrolled students as part of a departmental initiative.

Academic Tutor

STEM Topics, Private Clients

Various Locations

2018-Present

Worked one-on-one with students of high-school and college age on topics ranging from high school chemistry to Physics Olympiad preparation.

Mentorship

Ashleigh Smith

Undergraduate Mentee, Anomalous Diffusion

Chapel Hill, NC

2024-Present

Ashleigh created a PAAm gel protocol to meet complex experimental requirements for a poroelastic substrate in an isolated chamber. As she has matured as a researcher, this work has segued into a more autonomous project studying the wicking effect of poroelastic media.

Mason Miller

Undergraduate Mentee, Anomalous Diffusion

Chapel Hill, NC

2025-Present

Mason designed a 3D printed strain control system and ran many strain experiments in poroelastic media. Mason is now working to characterize the rheologic properties of poroelastic media during strains via atomic force microscopy.

Bradley Mankus

Undergraduate Mentee, Anomalous Diffusion

Chapel Hill, NC

2025-Present

Bradley has improved upon the PAAm gel preparation protocol, has also begun work on developing novel microfluidics channels for testing the effect of advective flows on diffusion in poroelastic media.

Shivam Patel <i>Undergraduate Mentee, Anomalous Diffusion</i> Shivam has worked to expand my data analysis codebase, and is currently working to simulate strain-induced advection in poroelastic media.	Chapel Hill, NC 2025-Present
Kautilya Chopra <i>Undergraduate Mentee, Anomalous Diffusion</i> Kautilya created python-based demos of brownian motion from first principles. He then expanded the simulation to model particle trapping and other dynamics typical of diffusion in poroelastic media.	Chapel Hill, NC Summer 2025
Akshay Elango <i>Undergraduate Mentee, Low-Resource Microscopy</i> Akshay built a 3-D printed transmission microscope based on the OpenFlexure system. His reserch is currenty on hiatus while he works in an industrial internship, but he will return upon its completion.	Chapel Hill, NC 2023-Present
Anya Frazer <i>Undergraduate Mentee, Low-Resource Microscopy</i> Anya expanded upon the OpenFlexure Microscope system, enhancing the capabilities and software of the fluorecence scope, including upgrading it to conduct single particle tracking of nanoscopic particles. She later developed a 3-D printed molecular stamper designed to manufacture Janus particles at dramtically reduced cost in low-resource settings.	Chapel Hill, NC 2024-2025
Eric Chen <i>Undergraduate Mentee, Low-Resource Microscopy</i> Eric built a 3-D printed fluorecence microscope based on the OpenFlexure system. He also designed and manufactured his own filtersets for this system from bulk sheets of Schottglass to minimize costs to advance the low-resource intensity goals of the project.	Chapel Hill, NC 2023-2024
Joshua Adekunle <i>Undergraduate Mentee, Low-Resource Microscopy</i> Josh built a 3-D printed low-cost transmission microscope based on OpenFlexure.	Chapel Hill, NC 2023-2024

Outreach

Open Flexure Microscope and Molecular Stamper <i>UNC-Chapel Hill</i> Initiatives planned for building microscopes in low-resource highschoools in Eastern NC.	Chapel Hill, NC 2023-Present
North Carolina Science Festival <i>Morehead Planetarium and Science Center</i> Each year, during this public outreach event the Superfine research group conducts science demos and research talks targeted at general audiences.	Chapel Hill, NC 2023-Present
Hospital Volunteer <i>Wake Forest Baptist Medical Center</i> Volunteered in various patient-facing positions for over 300 hours.	Winston-Salem, NC 2015-2017

Languages

English: Fluent	<i>Native Speaker</i>
Modern Greek: Intermediate	<i>Conversational speaker.</i>
Classical Latin: Intermediate	<i>Seven years of formal instruction</i>